

1.7 Quiz: Measurement

Reference (provided). average speed = $\frac{\text{distance}}{\text{time}}$

$$1 \text{ m} \approx 3.28 \text{ ft} \quad 1 \text{ mi} = 5280 \text{ ft} \quad 1 \text{ km} = 1000 \text{ m} \quad 1 \text{ m} = 100 \text{ cm} \quad 1 \text{ cm} = 10 \text{ mm}$$

$$1 \text{ h} = 3600 \text{ s} \quad 1 \text{ min} = 60 \text{ s} \quad 1 \text{ gal} \approx 3.785 \text{ L}$$

1. Write down the number of significant digits in each measurement.

(a) 0.0350 m 3 (c) 7.0 s 2 (e) 0.000 09 kg 1

(b) 806 g 3 (d) 12.50 cm 4 (f) 6.02×10^{23} 3

2. Round each value to three significant figures. Copy the calculator display followed by three dots, then round: $\sqrt{2} = 1.4142135 \dots \approx 1.41$

(a) $\sqrt{13} = 3.605551 \dots$
 ≈ 3.61

(b) The height of Mount Everest,
8848.86 m
8850 m

or
 $8.85 \times 10^3 \text{ m}$

3. Write in scientific notation, rounded to three significant figures.

(a) The distance light travels in one year, about 9,460,730,000,000 km (b) The charge of an electron, 0.000 000 000 000 000 000 160 2 C

$\approx 9.46 \times 10^{12} \text{ km}$

$1.60 \times 10^{-19} \text{ C}$

4. Write each value as an ordinary decimal number.

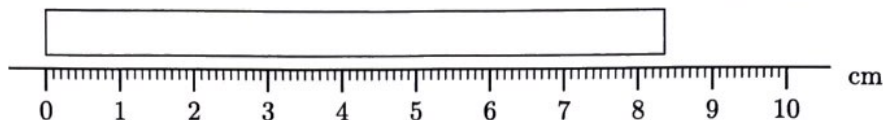
(a) The width of a human hair,
 $7.0 \times 10^{-5} \text{ m}$

0.000 070 m

(b) The closest distance from the Earth to Mars, $5.46 \times 10^7 \text{ km}$

54,600,000 km

5. The bar below is measured with a centimeter ruler marked in millimeters.



- (a) Record the length of the bar. Include one estimated digit — the last digit, the one you are not certain of.

8.36 cm

- (b) How many significant figures does your measurement have?

3

- (c) Write your measurement in millimeters.

83.6 mm

6. Convert between units, rounding to three significant figures. Write each conversion as a single chain of fractions and show the units cancelling.

- (a) Taipei 101 is 508 meters tall. What is its height in feet?

$$508 \text{ m} \times \frac{3.28 \text{ ft}}{1 \text{ m}} = 1666.24 \text{ ft} \\ \approx 1670 \text{ ft.}$$

- (b) Spain's AVE high-speed train runs at 310 kilometers per hour. What is this speed in meters per second?

$$310 \frac{\text{km}}{\text{hr}} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ hr}}{3600 \text{ s}} = 86.111... \\ \approx 86.1 \text{ m/s} \\ (\text{or } 86 \text{ m/s})$$

7. A cheetah has been timed running at 29 meters per second. What is this speed in miles per hour?

$$29 \text{ m/s} \times \frac{3.28 \text{ ft}}{1 \text{ m}} \times \frac{1 \text{ mile}}{5280 \text{ ft}} \times \frac{3600 \text{ s}}{1 \text{ hr}} = 64.854... \\ \approx 65 \text{ mph}$$

Early finisher

8. Mars is 5.46×10^7 km from the Earth at its closest. A radio command sent to a rover on Mars travels at the speed of light, 3.00×10^8 m/s. How many minutes does the command take to arrive?

$$\begin{aligned} d &= 5.46 \times 10^7 \text{ km} \\ r &= 3.00 \times 10^8 \text{ m/s} \\ &= 3.00 \times 10^5 \text{ km/s} \\ t &= \frac{5.46 \times 10^7 \text{ km}}{3.00 \times 10^5 \text{ km/s}} \end{aligned}$$

$$= 182 \text{ seconds}$$

$$\begin{aligned} 182 \text{ s} \times \frac{1 \text{ min}}{60 \text{ s}} &= 3.033 \text{ min} \\ &\approx 3.03 \text{ minutes} \end{aligned}$$

$$r \cdot t = d$$

$$\frac{d}{t} = r$$

$$\frac{d}{r} = t$$

9. The Atlantic Ocean widens by about 2.5 centimeters per year. Columbus first crossed it in 1492. By how many meters has the Atlantic widened between 1492 and 2026?

$$t = 2026 - 1492 = 534 \text{ years}$$

$$d = 2.5 \frac{\text{cm}}{\text{y}} \times 534 \text{ y}$$

$$= 13.35 \text{ m}$$

$$\approx 13 \text{ meters}$$

rate

10. A US shower head lets through at most 2.5 gallons of water per minute. How many liters of water does an 8.0-minute shower use?

$$\text{Quantity} = 2.5 \frac{\text{gallons}}{\text{minute}} \times 8.0 \text{ minutes}$$

$$= 20 \text{ gallons}$$

$$20 \text{ gallons} \times \frac{3.785 \text{ L}}{1 \text{ gal}} = 75.7 \text{ L} \\ \approx 76 \text{ liters}$$

rate

11. New York City uses about 1.0 billion gallons of water per day, for a population of about 8.5 million people. How many liters of water is that per person per day? Round your answer to the number of significant figures the data supports.

$$\frac{\text{gal}}{\text{person} \cdot \text{day}} = \frac{1.0 \times 10^9 \text{ gal/day}}{8.5 \times 10^6}$$

$$= 117.647... \text{ gallons/p.d.}$$

$$117.647... \frac{\text{g}}{\text{p.d.}} \times \frac{3.785 \text{ L}}{1 \text{ gallon}} = 445.294... \frac{\text{L}}{\text{p.d.}} \\ \approx 450 \frac{\text{L}}{\text{p.d.}}$$

(2 sig figs)